
Nomenclature and Context Sensitivity of Zero-Shot Biological Vision–Language Models for Bangladeshi Freshwater Fish Recognition

Anonymous Author(s)

Abstract

1 Zero-shot vision–language models (VLMs) are increasingly used as training-free species
2 recognizers, but reported accuracy can reflect more than visual species knowledge. We
3 audit CLIP, BioCLIP, BioCLIP2, and a multilingual Jina CLIP v2 control on seven
4 freshwater-fish categories from two Bangladeshi sources (10,321 images). BioCLIP2
5 reaches 72.36% on BFF-15 with English common names and 68.91% on SylFishBD
6 with scientific names, versus 25.15% and 14.40% for generic CLIP. BioCLIP2 Bengali
7 prompts are near chance in balanced accuracy (14.22–14.29%); Jina partially recovers
8 Bengali discrimination to 21.89% and 16.36%, but bare Bengali names return to 14.29%
9 on both sources. Paired SylFishBD interventions show no significant weak-blur effect,
10 modest losses from stronger blur/gray masking, a larger white-mask artifact, and strong
11 species dependence. Zero-shot biological VLM scores therefore jointly reflect biological
12 specialization, multilingual alignment, nomenclature, prompt formulation, and context.

13 1 Introduction

14 Automated species recognition supports biodiversity monitoring, fisheries, aquaculture, and market
15 inspection. Fish recognition is difficult because related taxa may differ subtly while pose, illumination,
16 occlusion, and background vary. Conventional systems therefore rely on labeled data and supervised
17 transfer learning [13, 19, 4], but high in-domain accuracy does not establish transferable biological
18 knowledge.

19 Contrastive vision–language pre-training offers a training-free alternative. CLIP enables zero-shot
20 image–text classification [12], while BioCLIP and BioCLIP2 add biologically structured pre-training
21 for fine-grained recognition [15, 7]. Fish work has also explored language supervision and training-
22 free multimodal recognition [3, 8].

23 For VLMs, the class label is part of the classifier: small wording changes can alter CLIP predictions
24 [21, 20]. Biology amplifies this issue because one organism may have vernacular, English-common,
25 scientific, historical, or synonymous names; scientific binomials are not always optimal [11], and
26 multilingual VLMs show large cross-language gaps [5, 2]. In Bangladesh, success through English
27 or Latin strings but failure through Bengali names can overstate local accessibility.

28 Visual context introduces a parallel ambiguity: classifiers can exploit backgrounds and shortcuts
29 [6, 18], while CLIP can retain domain-sensitive information [17]. We therefore ask: *how stable*
30 *is zero-shot biological recognition when the model, language, nomenclature, prompt, or context*
31 *changes?*

32 Our contributions are:

- 33 • a cross-model, cross-source zero-shot benchmark of CLIP, BioCLIP, and BioCLIP2 on seven
34 Bangladeshi freshwater-fish categories;

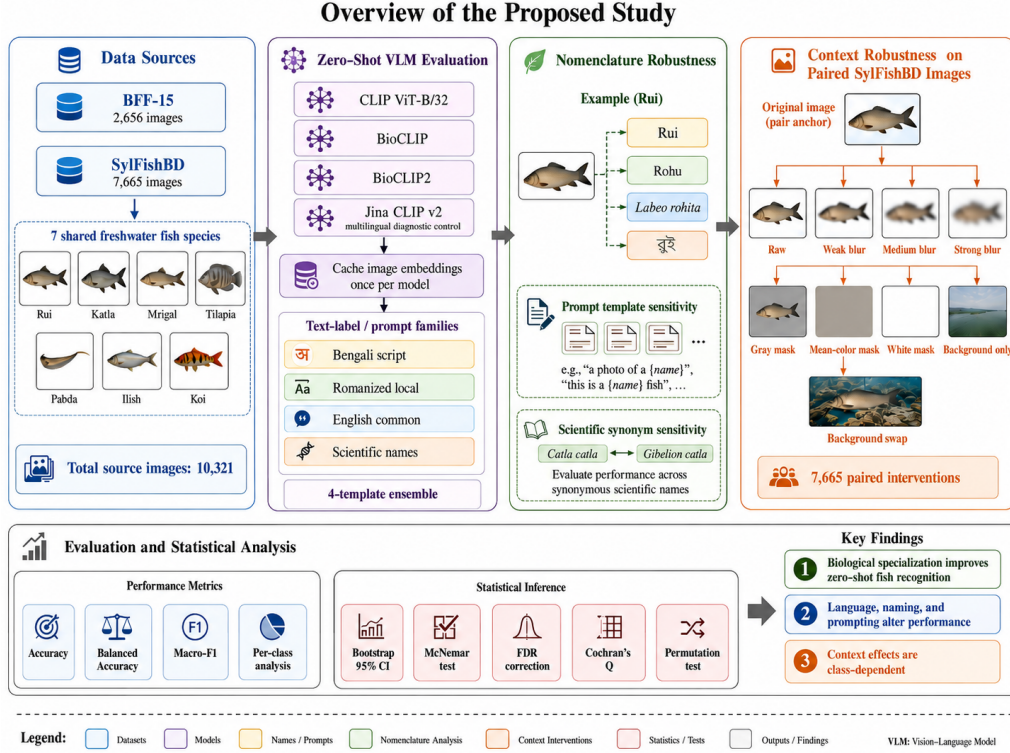


Figure 1: Study overview. Seven shared fish categories are evaluated with frozen CLIP, BioCLIP, and BioCLIP2; Jina CLIP v2 is a multilingual Bengali-interface control. We probe nomenclature/prompt robustness, paired context interventions, and statistical reliability.

- a nomenclature audit spanning Romanized vernacular, Bengali script, English-common and scientific names, prompt templates, and scientific synonyms, augmented with a multilingual Jina CLIP v2 control that separates multilingual alignment from biological specialization;
- paired context interventions on 7,665 SylFishBD images using graded blur, neutral masks, cropping, background-only views, and cross-species background swaps, with paired inference and multiple-testing correction.

2 Related Work

Prompt and nomenclature robustness. CLIP generates class prototypes through language rather than fixed task weights [12]. CoOp/CoCoOp showed that prompt context changes downstream behavior [21, 20]; in species recognition, common English names can outperform scientific names [11]. We keep weights fixed and treat name/prompt choices as evaluation variables.

Biological and multilingual VLMs. BioCLIP and BioCLIP2 use taxonomically informed contrastive pre-training at increasing scale [15, 7]. Multilingual evaluation reveals a different axis of quality: Babel-ImageNet reports large cross-language variation [5], while uCLIP extends VLMs multilingually [2]. We use multilingual Jina CLIP v2 [10] as a Bengali-alignment control, not a biological baseline.

Regional fish data and context. Bangladesh-specific fish resources include SylFishBD with SAM-derived instance masks [1, 9] and BFF-15 [16]. Recent self-supervised work achieves high in-domain accuracy after fine-tuning [14]; our models and prompts remain frozen. Background controls are motivated by evidence that context alone can retain class information [18].

Table 1: Shared seven-category benchmark. SylFishBD counts refer to source images; each has a paired segmentation mask.

Class	Romanized	English	Scientific	BFF	Syl
Rui	Rui	Rohu	<i>L. rohita</i>	514	1670
Katla	Katla	Catla	<i>C. catla</i>	427	1133
Mrigal	Mrigal	Mrigal carp	<i>C. cirrhosus</i>	317	1293
Tilapia	Tilapia	Nile tilapia	<i>O. niloticus</i>	383	1326
Pabda	Pabda	Pabda catfish	<i>O. pabda</i>	348	862
Ilish	Ilish	Hilsa	<i>T. ilisha</i>	233	789
Koi	Koi	Climbing perch	<i>A. testudineus</i>	434	592
Total				2656	7665

3 Methodology

3.1 Data harmonization

We use seven categories shared by BFF-15 and SylFishBD: Rui, Katla, Mrigal, Tilapia, Pabda, Ilish, and Koi. BFF-15 contributes 2,656 images and SylFishBD 7,665 (10,321 total); every selected SylFishBD image has a binary mask. Table 1 gives counts/nomenclature. All models remain frozen, so no train/validation/test split is required.

SHA-256 found no exact cross-source duplicates; a 64-bit perceptual-hash search flagged 73 low-distance pairs (≤ 6 bits) for review. Because similar fish can yield similar hashes, we flag rather than automatically exclude them and describe the collections as two public sources, not fully independent sources.

3.2 Frozen zero-shot classification and prompt construction

We evaluate CLIP ViT-B/32 (OpenAI), BioCLIP, and BioCLIP2. For normalized image embedding $f_I(x)$ and class name n_c , we instantiate four templates—"a photo/image/photograph/specimen of n_c , a fish species"—normalize each text embedding, and form the normalized mean prototype

$$t_c = \frac{\frac{1}{K} \sum_{k=1}^K \hat{f}_T(p_k(n_c))}{\left\| \frac{1}{K} \sum_{k=1}^K \hat{f}_T(p_k(n_c)) \right\|_2}, \quad K = 4. \quad (1)$$

Prediction is $\hat{y} = \arg \max_c f_I(x)^\top t_c$. Cached image embeddings ensure nomenclature experiments change only the text-side classifier.

We test Romanized, English-common, and scientific names, plus Bengali script for BioCLIP2. Templates are also evaluated individually. Scientific-name controls substitute *Catla catla* with *Gibelion catla* or *Labeo catla*, and *Cirrhinus cirrhosus* with *C. mrigala*; these are nomenclatural variants, not claims of uniquely correct taxonomy.

3.3 Multilingual Bengali control

To test whether BioCLIP2’s Bengali failure reflects text alignment, we add frozen Jina CLIP v2 [10]. The same 10,321 images/classes use Bengali, Romanized, English-common, and scientific labels with the same four-template ensemble and a 512-dimensional Matryoshka-truncated representation; no benchmark tuning is performed.

A secondary *name-only* control embeds only Bengali class names, testing whether multilingual support alone suffices or interacts with prompt framing. This directly tests whether BioCLIP2’s near-chance Bengali result is partly a text-alignment limitation rather than a property of Bengali nomenclature itself. Jina is not used for context interventions; it is diagnostic for multilingual alignment versus biological specialization.

3.4 Paired context interventions

SylFishBD masks preserve foreground pixels while altering background. We apply Gaussian blur ($\sigma = 0.01, 0.03, 0.06$ times the shorter dimension), white/gray/mean-color masks, a 10%-expanded

Table 2: Primary zero-shot benchmark. Values are accuracy / macro-F1 (%).

Model	Prompt	BFF-15	SylFishBD
CLIP ViT-B/32	Romanized	10.96 / 5.64	16.95 / 13.15
	English	25.15 / 20.03	23.80 / 22.05
	Scientific	15.40 / 6.97	14.40 / 7.21
BioCLIP	Romanized	25.26 / 18.28	27.93 / 16.63
	English	54.89 / 51.61	52.26 / 46.64
	Scientific	47.82 / 43.34	58.88 / 57.45
BioCLIP2	Romanized	35.77 / 25.98	37.56 / 25.19
	English	72.36 / 67.33	64.59 / 64.30
	Scientific	69.99 / 68.85	68.91 / 69.68

88 tight crop, foreground removal with Telea inpainting, and cross-species background swaps. These are
89 stress tests, not perfect causal isolation, because synthetic transformations introduce distribution shift.

90 3.5 Metrics and inference

91 We report accuracy, balanced accuracy, macro-F1, and per-class effects. Class-stratified bootstrap
92 CIs use 1,000 resamples for the primary benchmark and 2,000 for robustness/multilingual controls.
93 Paired comparisons use exact two-sided McNemar tests and bootstrap accuracy differences with
94 Benjamini–Hochberg FDR correction. Cochran’s Q tests context conditions; donor-follow uses 5,000
95 label permutations. Seed is 42.

96 4 Results and Discussion

97 4.1 Biology-specialized VLMs outperform the tested generic CLIP baseline

98 Table 2 shows a large gap between generic and biology-specialized VLMs. With scientific names,
99 CLIP scores 15.40%/14.40% on BFF-15/SylFishBD, BioCLIP 47.82%/58.88%, and BioCLIP2
100 69.99%/68.91%. With English names BioCLIP2 reaches 72.36%/64.59%. Relative to BioCLIP,
101 BioCLIP2 gains 22.18 and 10.03 points under scientific prompting (paired 95% CIs [19.73, 24.66]
102 and [8.91, 11.17], both $q < 0.001$). Because checkpoints also differ in scale/training recipe, we do
103 not attribute the full gap solely to specialization.

104 4.2 Nomenclature and prompt formulation are part of the classifier

105 With identical BioCLIP2 image embeddings, BFF-15 scores 35.77%/72.36%/69.99% for Roman-
106 ized/English/scientific names; SylFishBD scores 37.56%/64.59%/68.91%. Scientific terminology is
107 therefore not universally optimal.

108 Bengali prompts yield 16.27%/7.74% accuracy and 14.22%/14.29% balanced accuracy, essentially
109 seven-class chance (14.29%). Thus these Bengali fish-name prompts provide little discrimination for
110 the tested BioCLIP2 text encoder.

111 Across four individual scientific templates, BioCLIP2 spans 56.06–77.07% on BFF-15 and 61.44–
112 79.71% on SylFishBD. Replacing *Catla catla* by *Gibelion catla* raises SylFishBD accuracy 5.14
113 points ($q < 0.001$), while *Labeo catla* lowers it 3.46 points and lowers BFF-15 by 8.58 points.
114 Replacing *Cirrhinus cirrhosus* by *C. mrigala* improves both sources by about two points. Thus
115 “scientific prompt” is not fully specified without the exact string/template.

116 4.3 Multilingual alignment partially recovers Bengali, but does not replace biological 117 specialization

118 Under the same four-template ensemble, Jina CLIP v2 reaches 21.89% Bengali balanced accuracy
119 on BFF-15 (95% CI [20.77, 23.01]) and 16.36% on SylFishBD ([15.64, 17.05]), versus BioCLIP2’s
120 14.22%/14.29%. This descriptive control suggests that the near-chance BioCLIP2 result is partly
121 model/interface-specific.

122 Recovery remains incomplete: English exceeds Bengali accuracy by 4.10 points on BFF-15 (CI [2.86,
123 5.35], $q < 10^{-8}$) and 13.70 on SylFishBD ([12.77, 14.66], $q < 10^{-98}$). Bengali and Romanized
124 are indistinguishable on BFF-15 ($q = 0.617$), while Romanized exceeds Bengali by 4.66 points on
125 SylFishBD ($q < 10^{-19}$).

126 Jina’s English accuracy (29.41%/23.78%) remains far below BioCLIP2 (72.36%/64.59%), so multi-
127 lingual alignment does not replace biological specialization. Bare Bengali names also collapse to
128 exactly 14.29% balanced accuracy on both sources, showing an interaction with prompt framing.
129 Together, the controls expose a trade-off: stronger multilingual alignment can recover Bengali signal
130 without matching BioCLIP2’s fine-grained biological accuracy. We therefore interpret Jina as an
131 interface diagnostic, not evidence about Bengali understanding in general.

Table 3: Jina CLIP v2 multilingual control. Values are accuracy / balanced accuracy / macro-F1 (%).

Prompt family	BFF-15	SylFishBD
Bengali script	25.30 / 21.89 / 16.31	10.08 / 16.36 / 6.04
Romanized	24.89 / 23.93 / 19.05	14.74 / 14.78 / 12.06
English common	29.41 / 28.43 / 17.87	23.78 / 18.01 / 15.62
Scientific	16.11 / 15.96 / 13.56	9.13 / 10.27 / 9.75

132 4.4 Context sensitivity is real but intervention-dependent

133 The white-mask comparison drops BioCLIP2 from 68.91% to 60.52% (−8.39 points), but the
134 intervention ladder qualifies this effect (Table 4). Weak blur is nonsignificant (−0.47; CI [−1.34,
135 0.46], $q = 0.347$); medium/strong blur reduce accuracy 2.18/2.87 points and gray/mean-color
136 masking about 3.8 points (all $q < 0.001$). Thus white masking adds substantial distribution shift.

Table 4: BioCLIP2 context ladder on 7,665 paired SylFishBD images. Δ is change from raw; brackets are paired-bootstrap 95% CIs.

Condition	Acc.	Δ (pp)	FDR q
Raw	68.91	—	—
Weak blur	68.44	−0.47 [−1.34, 0.46]	0.347
Medium blur	66.73	−2.18 [−3.11, −1.23]	< 0.001
Strong blur	66.04	−2.87 [−3.81, −1.94]	< 0.001
Gray mask	65.11	−3.80 [−4.79, −2.83]	< 0.001
Mean-color mask	65.10	−3.81 [−4.76, −2.81]	< 0.001
White mask	60.52	−8.39 [−9.56, −7.25]	< 0.001

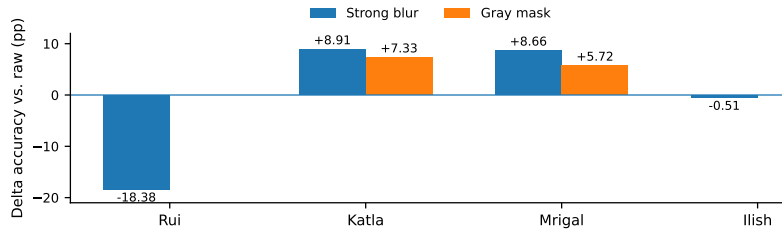


Figure 2: Class-conditional changes from raw SylFishBD. Strong blur helps Katla/Mrigal but sharply hurts Rui; Ilish is nearly unchanged. Gray-mask values are shown where exact effects are available.

137 Aggregate means conceal class dependence (Fig. 2). Strong blur improves Katla/Mrigal by 8.91/8.66
138 points, hurts Rui by 18.38, and barely changes Ilish (−0.51). Gray masking shows the same
139 heterogeneity: Katla/Mrigal improve (+7.33/+5.72), while Rui, Koi, Tilapia, and Pabda decline.

140 Supporting controls reject simple background-label explanations. Background-only images retain
141 17.90% accuracy (21.48% balanced; −51.01 points, CI [−52.17, −49.84]), confirming dominant
142 foreground information. Cross-species swaps reduce accuracy to 56.76% (−12.15; CI [−13.24,
143 −11.06]), yet donor-follow is only 8.88% versus a 15.10% permutation-null mean. Tight cropping
144 reaches 55.15%, and five mask-geometry properties are nonsignificant after FDR ($q \geq 0.097$).

4.5 Implications and limitations

Three implications follow. Biology-specialized VLMs outperform the tested generic CLIP baseline, although scale/training recipe also differ. Zero-shot accuracy is not an image-encoder property alone: language, exact taxonomic string, and prompt frame instantiate the classifier, as the multilingual/name-only controls demonstrate. Context robustness likewise requires a ladder of interventions rather than one synthetic mask. Practically, reporting only the best English/scientific prompt can overstate accessibility when a strong visual model exposes a weak native-language interface.

Limitations include only seven categories/two sources, unknown pre-training exposure to related imagery, one multilingual control with a 512-D truncated representation, Bengali results specific to the tested names/templates, and synonym tests limited to Katla/Mrigal. One multilingual model cannot establish a general property of multilingual VLMs. Context interventions change image distribution as well as information content, so conclusions rely on agreement across blur, masks, crop, background-only, and swap controls rather than perfect causal decomposition.

5 Conclusion

We audit zero-shot freshwater-fish recognition across generic, biology-specialized, and multilingual VLM interfaces. BioCLIP2 is strongest overall but remains sensitive to nomenclature, language, prompt syntax, synonyms, and context. Jina partially recovers Bengali discrimination yet is weaker at fine-grained recognition and collapses to chance with bare Bengali names. Paired context interventions show small average effects for mild degradation, larger synthetic-mask artifacts, and strong species dependence. Biological specialization and multilingual alignment address different failure modes; benchmarks should therefore report and stress-test language, exact nomenclature, prompt templates, and context.

References

- [1] Shakib Absar, I. Ahmed, M. I. Khaliq, M. Shorfuazzaman, M. M. Rahman, A. U. Haque, and M. S. R. Kohinoor. A freshwater fish dataset for visual recognition with manually localized rois and sam-derived instance masks. *Scientific Data*, 2026.
- [2] D. Chung, D. Shin, Y. Sung, S. Moon, J. Jeon, and B.-J. Lee. uclip: Parameter-efficient multilingual extension of vision-language models with unpaired data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 3399–3406, 2026.
- [3] K. Dai, J. Shao, B. Gong, L. Jing, and Y. Chen. Clip-fssc: A transferable visual model for fish and shrimp species classification based on natural language supervision. *Aquacultural Engineering*, 107:102460, 2024.
- [4] P. K. Das, M. A. Kawsar, P. B. Paul, M. A. A. M. Hridoy, M. S. Hossain, and S. Niloy. Bd-freshwater-fish: An image dataset from bangladesh for ai-powered automatic fish species classification and detection toward smart aquaculture. *Data in Brief*, 57:111132, 2024.
- [5] Gregor Geigle, Radu Timofte, and Goran Glavas. Babel-imagenet: Massively multilingual evaluation of vision-and-language representations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5064–5084, 2024.
- [6] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.
- [7] Jian Gu, Samuel Stevens, Elizabeth G. Campolongo, Matthew J. Thompson, Ning Zhang, Jiaman Wu, Alex Kopanev, Zixuan Mai, Andrew E. White, James Balhoff, Wasila Dahdul, David Rubenstein, Hilmar Lapp, Tanya Berger-Wolf, Wei-Lun Chao, and Yu Su. Bioclip 2: Emergent properties from scaling hierarchical contrastive learning. In *Advances in Neural Information Processing Systems*, volume 38, pages 113880–113913, 2025.
- [8] J. A. Jose and R. Thomas. Zero shot multistage hierarchical fish classification using yoloe and large language model. *Computers and Electronics in Agriculture*, 242:111288, 2026.

- [9] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023.
- [10] Andreas Koukounas, Georgios Mastrapas, Bo Wang, Mohammad Kalim Akram, Sedigheh Eslami, Michael Günther, Isabelle Mohr, Saba Sturua, Scott Martens, Nan Wang, and Han Xiao. jina-clip-v2: Multilingual multimodal embeddings for text and images, 2024.
- [11] Shubham Parashar, Zhiqiu Lin, Yanan Li, and Shu Kong. Prompting scientific names for zero-shot species recognition. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9856–9861, 2023.
- [12] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763, 2021.
- [13] Syed Zain Hassan Shah, Hafiz Tayyab Rauf, Muhammad IkramUllah, Muhammad Salman Khalid, Muhammad Farooq, Maryam Fatima, and Syed Ahmad Chan Bukhari. Fish-pak: Fish species dataset from pakistan for visual features based classification. *Data in Brief*, 27:104565, 2019.
- [14] S. N. Siam, M. S. Hasan, T. S. Raita, S. S. Israt, A. Hossain, M. S. H. Khan, R. A. Tuhin, M. R. A. Rashid, R. U. Islam, S. H. Ripon, A. W. Reza, and M. S. Hossain. Freshwater fish species classification using self supervised learning. In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, pages 3458–3463. IEEE, 2025.
- [15] Samuel Stevens, Jiaman Wu, Matthew J. Thompson, Elizabeth G. Campolongo, Chan Hee Song, David E. Carlyn, Li Dong, Wasila M. Dahdul, Charles Stewart, Tanya Berger-Wolf, Wei-Lun Chao, and Yu Su. Bioclip: A vision foundation model for the tree of life. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19412–19424, 2024.
- [16] TheShahidul. Bangladeshi freshwater fish image dataset (bff-15). Kaggle dataset, 2026. Accessed Aug. 25, 2026.
- [17] C. Wen, Z. Peng, Y. Huang, X. Yang, and W. Shen. Domain generalization in clip via learning with diverse text prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9559–9569, 2025.
- [18] Kai Yuan Xiao, Logan Engstrom, Andrew Ilyas, and Aleksander Madry. Noise or signal: The role of image backgrounds in object recognition. In *International Conference on Learning Representations*, 2021.
- [19] Xiaoqian Xu, Wei Li, and Qing Duan. Transfer learning and se-resnet152 networks-based for small-scale unbalanced fish species identification. *Computers and Electronics in Agriculture*, 180:105878, 2021.
- [20] Kaiyang Zhou, Jingkan Yang, Chen Change Loy, and Ziwei Liu. Conditional prompt learning for vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16816–16825, 2022.
- [21] Kaiyang Zhou, Jingkan Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9):2337–2348, 2022.